

# Nicole Ye's Portfolio

[Linkedin](#) | [Nicole Ye's Resume](#) | [My Website!](#) | [n7ye@uwaterloo.ca](mailto:n7ye@uwaterloo.ca) | [Github](#)

**If you have any inquiries, please don't hesitate to reach out!**

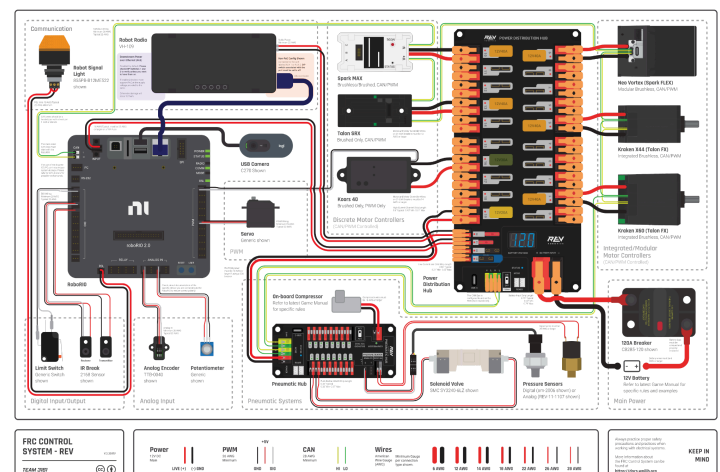
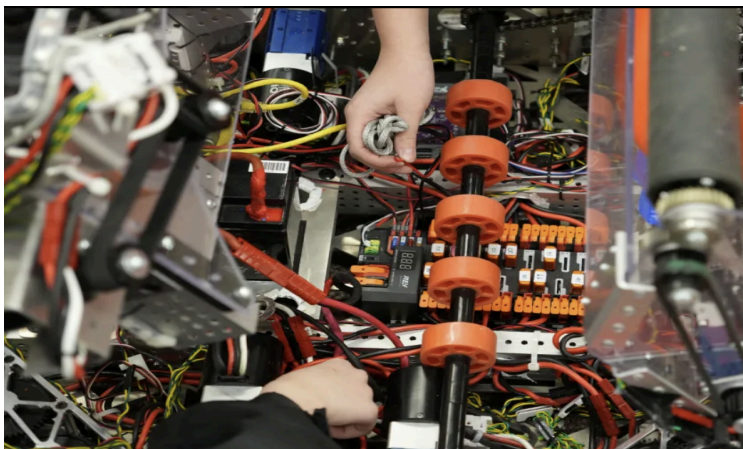
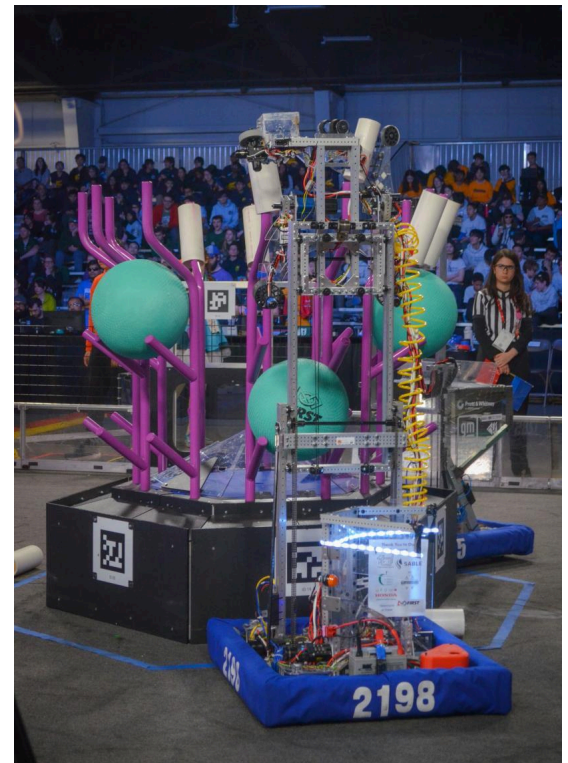
## SKILLS

- **Firmware & Embedded:** ESP32, Raspberry Pi Pico, C++, Java, PlatformIO, I2C, UART, SPI
- **Electronics:** KiCad, EasyEDA, PCB Schematic & Layout, Soldering, Multimeter, Oscilloscope, Power Supply, Cable Management, Crimping
- **Sensors & Protocols:** VL53L4CX ToF, RDM630 RFID, Hall-Effect, Encoders,
- **Mechanical/CAD:** Onshape, SolidWorks, BambuStudio, 3-D Printing, Sheet Metal, Snap-Fits

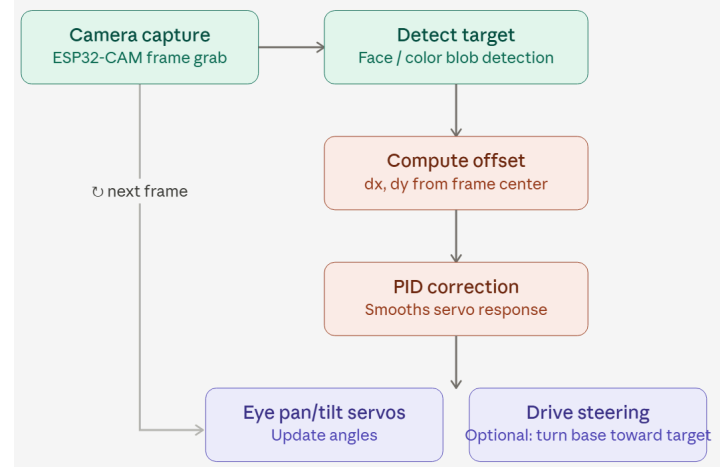
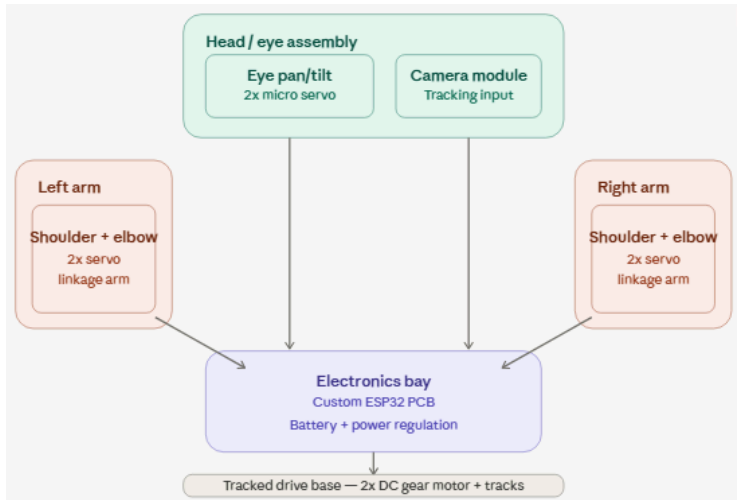
# Team 2198 Paradigm Shift Robotics Electrical System

Sep. 2021 - June 2025

- Utilized [TKO Circuit Simulator](#) to design the layout of electrical components 2024-2025 robot
- Designed and implemented a customized electrical system, including wiring, power distribution, and sensor integration, ensuring compliance with competition standards
- Integrated a **line sensor** to detect whether a game piece was secured in the holding mechanism, improving automation
- Wired **encoders** for precise motor control and accurate movement tracking across drive and manipulator systems
- Implemented **color sensors** to differentiate game piece types and enable autonomous decision-making
- Programmed AprilTag-based field-relative localization in Java using Limelight vision co-processor, enabling automated robot alignment to scoring positions and reducing driver repositioning time during matches



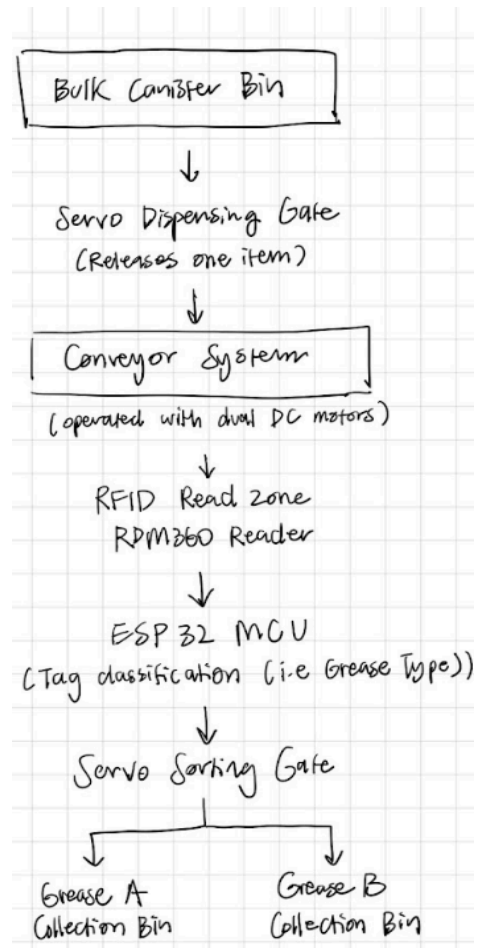
# WALL-E Animatronic Robot (In Progress)



- Designed a custom 2-layer ESP32-based PCB in KiCad integrating USB programming, 3.3V/5V power regulation, a DRV8833 motor driver, connectors for 6 SG90 servos, dual N20 gear motors, and an ESP32-CAM vision module.
- Applied mixed-signal PCB design practices including RF antenna keep-outs, power/logic separation, decoupling networks, bulk capacitance for transient loads, and DRC/manufacturability verification prior to fabrication.
- Performed board bring-up and hardware validation by verifying power rails, flashing firmware through an onboard USB-to-serial interface, and debugging servo and motor outputs using oscilloscopes and multimeters.
- Developed ESP32 firmware enabling Wi-Fi teleoperation through a browser-based interface, implementing real-time control of six servos and differential-drive motors using LEDC PWM and WebSocket/HTTP communication.
- Designed and assembled a fully articulated 3D-printed WALL-E platform featuring tracked locomotion, dual-arm actuation, pan-tilt eyes, integrated battery power, and onboard electronics packaging.
- Implemented machine-vision tracking using an ESP32-CAM to perform color-blob detection, transmitting target offsets over UDP to the main controller for closed-loop servo tracking using proportional/PID control.
- Integrated manual and autonomous operating modes with status indication, battery monitoring, and system documentation including block diagrams, pin mappings, and communication interface specifications.

# Automated RFID Conveyor Sorting System

- Designed and built an automated conveyor-based sorting system capable of dispensing, identifying, and sorting RFID-tagged canisters without operator intervention.
- Developed a servo-actuated dispensing mechanism that released RFID-tagged canisters individually from a bulk storage hopper onto the conveyor, ensuring reliable single-item processing.
- Integrated an RDM630 RFID reader with an ESP32 to decode tag IDs in real time and classify canisters based on prefix patterns corresponding to grease type.
- Designed a dual-motor conveyor system to transport canisters through the sensing and sorting stages while maintaining consistent spacing and throughput.
- Implemented a servo-driven diverter gate that redirected each canister into the appropriate collection bin based on the RFID classification result.
- Developed embedded control logic to synchronize dispensing, tag acquisition, conveyor transport, and sorting actions into a deterministic processing sequence.
- Achieved continuous operation at approximately 720 tags per hour through a 5-second dispense-read-sort cycle while maintaining accurate classification performance.



# Robot Part Drawings OnShape/Solidworks

- Utilized Solidworks and Onshape to create DXF, STEP, and PDF drawings of sheet metal of First Robotics Robots (2021-2025) for CNC processing
- Designed and fabricated gussets for a FIRST Robotics robot, ensuring precise alignment and secure connections between major mechanisms to enhance structural integrity
- Followed set of tools utilizing standard views, section, and detail dimensions such as model items and smart Dimension, and annotations (including center marks and bill of materials)

